Comp 442 Progress Report

"Phish Detector"

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1. Introduction

1.1 Project Overview

The Phish Detector project aims to develop a machine learning-based system capable of identifying spam and phishing messages. The system will use supervised machine learning techniques and natural language processing methods to analyze message content and classify messages as either legitimate or malicious. By identifying suspicious patterns, keywords, and message characteristics, the project seeks to improve user protection against common cybersecurity threats such as phishing attacks, scams, and fraudulent communications.

1.2 Problem Definition and Motivation

Spam and phishing messages continue to be one of the most common cybersecurity threats affecting users today. These attacks often rely on deceptive wording, malicious links, and false urgency to trick users into revealing sensitive information or performing harmful actions. Traditional rule-based filtering systems can block some threats but often struggle to adapt to evolving attack patterns. This project aims to address this problem by applying machine learning techniques that can learn from historical examples and identify suspicious messages more effectively. The motivation behind this project is to demonstrate how machine learning can be used to improve online safety and reduce user exposure to phishing attacks.

2. Review of Previous Assignments

2.1 Assignment 1: Project Proposal

Problem definition: The project addresses the problem of identifying spam and phishing messages that can expose users to identity theft, financial fraud, malware, and privacy breaches. The goal is to create a system capable of automatically distinguishing malicious messages from legitimate communications.

Business objectives: The primary objective is to develop a machine learning-based spam and phishing detection system that can accurately classify messages while minimizing false positives and false negatives. Additional objectives include improving user awareness, reducing exposure to phishing attempts, and demonstrating a practical cybersecurity solution.

Dataset selection: The Email Spam Detection Dataset obtained from Kaggle was selected for the project. The dataset contains 5,572 labeled messages classified as either spam or ham. The labeled nature of the dataset makes it suitable for supervised machine learning and text classification tasks.

Machine learning goals: The machine learning goal is to develop a classification model capable of identifying spam and phishing messages based on message content. Multiple classification algorithms will be evaluated and compared using standard performance metrics such as accuracy, precision, recall, and F1-score.

2.2 Assignment 2: Data Exploration Using Statistical Descriptions

Dataset characteristics: The dataset contains 5,572 records and two attributes: Message and Category. The Category attribute contains binary labels indicating whether a message is spam or ham, while the Message attribute contains the text content used for analysis and classification.

Data quality findings: Analysis revealed that the dataset contains no missing values. However, 415 duplicate records were identified. Duplicate messages may introduce bias during training and will be removed during preprocessing.

Class distribution analysis: The dataset exhibits class imbalance. Ham messages account for 4,825 records (86.6%), while spam messages account for 747 records (13.4%). This imbalance may affect model performance and will be considered during evaluation.

Message length statistics: Statistical analysis of message length revealed a mean of 80.37 characters and a median of 61 characters. The distribution exhibits positive skewness and contains several outliers, indicating that while most messages are relatively short, some messages are substantially longer.

Word count statistics: Word count analysis showed an average of 15.58 words per message and a median of 12 words. Similar to message length, the distribution is positively skewed, with a small number of messages containing significantly more words than average.

Correlation analysis: A Pearson correlation analysis between Message Length and Word Count produced a correlation coefficient of 0.974, indicating a very strong positive relationship. Messages with more words generally have greater overall length.

2.3 Assignment 3: Data Exploration Using Visualization Techniques

Distribution visualizations: Several visualizations were created to examine the distribution of message characteristics, including histograms, KDE density plots, area plots, and boxplots. These visualizations confirmed the presence of positive skewness and outliers in both message length and word count distributions.

Class distribution visualizations: Vertical bar charts and pie charts were used to visualize the class distribution of spam and ham messages. These visualizations clearly demonstrated the class imbalance present within the dataset.

Feature relationship visualizations: Scatter plots, hexbin plots, and correlation heatmaps were used to explore relationships among numerical features. These visualizations confirmed the strong positive relationship between message length and word count.

Key findings: The analysis identified several important characteristics of the dataset, including class imbalance, duplicate records, positive skewness, outliers, and strong feature correlations. These findings provide valuable guidance for future preprocessing and model development activities.

3. Current Data Preprocessing Status

3.1 Data Cleaning

- Duplicate removal: A total of 415 duplicate records were identified during data exploration. These duplicate messages will be removed to prevent overrepresentation of specific message patterns during model training.
- Data quality improvements: Data quality assessment showed that the dataset contains no missing values. Additional cleaning steps will focus on removing duplicate records and ensuring consistent formatting of message content.

3.2 Data Transformation

- Lowercasing: All message text will be converted to lowercase to ensure consistent representation of words and reduce unnecessary feature duplication caused by capitalization differences.
- Punctuation removal: Punctuation marks, special characters, and non-essential symbols will be removed from messages to reduce noise and simplify text analysis.
- Tokenization: Tokenization will be applied to split messages into individual words or tokens. This step prepares the text for feature extraction techniques such as TF-IDF vectorization and keyword analysis.

3.3 Feature Selection

Message length was selected as a feature because it helps describe the structure of each message. The visualization analysis showed that most messages are short, but some messages are much longer than others. This means that message length may help the model recognize differences between messages.

Word count was also selected because it gives another way to measure message size. Messages with more words usually have longer message lengths. This means that word count may help the model recognize differences in message patterns.

URL count was selected because spam and phishing messages can contain links that try to direct users to outside websites. Counting the number of URLs in a message can help identify messages that are suspicious.

TF-IDF was selected because it can represent the importance of words in each message. The visualization analysis showed that spam messages often contained repeated words such as call, free, mobile, text, claim, prize, cash, and won. These word patterns can help the model learn which messages contain terms associated with spam.

Together, these selected features give the model structure-based information and text-based information. Message length, word count, and URL count describe the format of the message, while TF-IDF helps represent the actual words used.

3.4 Feature Extraction

Feature extraction is needed because the original message text is unstructured and cannot be used directly by most machine learning models. The message must be converted into numerical features before training.

The main feature extraction method planned for this project is TF-IDF vectorization. TF-IDF converts message text into numerical values based on how important a word is within a message compared to the full dataset. This is useful for spam detection because certain words appear more frequently in spam messages compared to ham messages.

The visualization analysis showed that spam messages frequently contained words such as call, free, mobile, text, claim, reply, prize, cash, and won. These repeated spam-related words support the use of TF-IDF because it allows the model to learn words associated with spam. Together, TF-IDF features and engineered text features will help the model learn the structure of spam and ham messages.

4. Current Training and Modeling Status

4.1 Planned Model Architecture

Several supervised machine learning algorithms are planned for implementation and comparison throughout the project. These models were selected because they are commonly used in text classification and spam detection tasks.

Naive Bayes is a probabilistic classification algorithm that is widely used for spam filtering and text classification. It performs well on text-based datasets and serves as an effective baseline model due to its simplicity and efficiency.

Logistic Regression is a supervised classification algorithm that predicts the probability that a message belongs to a particular class. It is commonly used in natural language processing applications and often performs well when combined with TF-IDF feature extraction.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. This model is capable of capturing more complex relationships within the data and will be compared against the simpler classification approaches.

The final model selected for deployment will be determined based on overall performance across multiple evaluation metrics.

4.2 Planned Training Setup

Training/test split: The dataset will be divided into separate training and testing sets to evaluate model performance on previously unseen data. A standard split such as 80% training and 20% testing will be considered, ensuring that both spam and ham messages remain proportionally represented in each subset.

Development environment: Model development, training, and evaluation will be performed using Python within Google Colab. Additional libraries such as Pandas, NLTK, and Scikit-learn will be used for data preprocessing, feature extraction, model training, and evaluation. GitHub will be used for collaboration and version control.

Evaluation metrics: Because the dataset contains class imbalance, model performance will be evaluated using multiple metrics rather than accuracy alone. Planned evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics will provide a more complete assessment of model performance and help identify whether the model is effectively detecting spam messages.

4.3 Regularization and Generalization Plan

Class weighting: Due to the class imbalance identified during data exploration, class weighting may be applied to reduce bias toward the majority class. This approach helps ensure that spam messages receive appropriate consideration during model training.
Cross validation: Cross-validation techniques will be used to assess model stability and generalization performance. By training and evaluating the model across multiple

subsets of the data, the team can obtain a more reliable estimate of real-world performance.
Feature selection: Feature selection techniques will be applied to identify the most useful attributes for classification. Features such as TF-IDF scores, message length, word count, URL count, and suspicious keyword frequency will be evaluated to determine their contribution to model performance. Reducing irrelevant features may improve accuracy, reduce overfitting, and decrease computational complexity.

5. Plan For Modeling, Evaluation, and Deployment

5.1 Planned Modeling Improvements

As the project progresses, several modeling improvements and experiments are planned to improve classification performance and generalization. The team will begin with baseline models such as Naive Bayes and Logistic Regression before evaluating more advanced approaches such as Random Forest.

One planned improvement is hyperparameter tuning to identify optimal model configurations. Parameters such as the number of trees in Random Forest, regularization strength in Logistic Regression, and feature selection thresholds will be adjusted and evaluated to improve performance.

Additional feature engineering techniques may also be explored. While the current feature set includes TF-IDF values, message length, word count, URL count, and suspicious keyword frequency, the team may experiment with n-grams and other text-based features to capture more contextual information from messages.

Model comparison will be a key part of the improvement process. Multiple algorithms will be trained and evaluated using the same dataset and performance metrics. The final model will be selected based on its ability to accurately classify spam messages while maintaining strong precision, recall, and F1-score performance on unseen data.

5.2 Evaluation Strategy

The model will be evaluated using multiple classification metrics instead of only accuracy. This is important because the dataset is imbalanced, containing more ham than spam messages. If accuracy is the only metric used, the model could appear to perform well but may not actually be well trained if it predicts most messages as ham.

The main evaluation metrics will include accuracy, precision, recall, F1-score, and a confusion matrix. Accuracy shows the overall percentage of correct predictions. Precision will determine how many messages predicted as spam were actually spam. Recall will show how many actual spam messages the model correctly identified. F1-score will provide a balance between precision and recall.

Recall is important for this project because missed spam messages are false negatives. In a phishing detection system, false negatives are risky because they are able to reach a user without warning. Precision is also important since too many false positives could incorrectly label legitimate messages as spam.

A confusion matrix will be used to show the number of true positives, true negatives, false positives, and false negatives. This will help the team understand not only how accurate the model is, but also what types of errors occur.

5.3 Deployment Plan

Streamlit web app: The final model will be deployed using Streamlit, a Python-based framework for creating interactive web applications. Streamlit was selected because it allows machine learning models to be integrated into a simple and user-friendly interface without requiring extensive web development experience. The application will serve as a demonstration of the spam and phishing detection system and allow users to interact directly with the trained model.

UI design: The planned user interface will allow users to enter or paste a message into a text input field. Once submitted, the application will process the message, apply the same preprocessing and feature extraction techniques used during training, and generate a prediction indicating whether the message is spam/phishing or legitimate.

The interface will display the classification result along with relevant information such as prediction confidence and warning messages when suspicious content is detected. The design will prioritize simplicity, usability, and clarity so that users can quickly understand the model's output. This deployment approach will provide a practical demonstration of how machine learning can be used to support cybersecurity awareness and spam detection.

6. Timeline

6.1 Task Table

Task	Team member	Status
Data cleaning	haley	IP
Feature extraction	patrick	planned
Model training	carlos	planned
Model eval	finn	planned
Deployment and demo	viet	planned

Final report and slides	Full team	planned

6.2 Gantt Chart

Task	Week 4	Week 5	Week 6
Data cleaning			
Feature extraction			
Model training			
Model eval			
Deployment and demo			
Final report and slides			

7. Roadblocks, Risks, and Contingency Plans

7.1 Current Roadblocks

There are several roadblocks that were identified during the data exploration and visualization phase. The first roadblock is class imbalance. The dataset contains many more ham messages than spam messages, which can cause the model to be biased toward ham messages. This can make the model less effective at detecting spam messages.

Another roadblock is the duplicate records that were found in the data. The dataset contains duplicate messages, which can cause the model to memorize repeated messages instead of learning the general spam and ham patterns.

The dataset also contains long-message outliers. Most messages are short, but some are much longer than the rest. The outliers may affect statistical analysis and feature distribution. However, they should not be removed automatically since they may contain useful information.

Text formatting is another challenge. Messages may include punctuation, capitalization differences, abbreviations, numbers, links, and unusual formatting. Since the data is unstructured text, these issues must be handled during preprocessing and feature extraction.

A final roadblock is that some features may provide overlapping information. An example of this is that message length and word count have a strong positive correlation. This means they are closely related and may not add unique information.

Comparison to other, larger datasets with 500K+ entries revealed (through confidence internals roughly ~70%) that the small data size is still consistent with the findings observed in larger datasets.

On the same note as the small dataset, our dataset is internationally based (with entries and characters showing it’s from the United Kingdom). Despite this challenge, it is for the most part consistent with more up to date findings from the larger Enron dataset containing 500,000 entries and dated for 2015-2016.

7.2 Risk Register

Risk	Likelihood	Impact	Contingency Plan
Class imbalance	High	High	Use precision, recall, F1-score, and confusion matrix instead of accuracy alone. Consider class weighting if needed
Duplicate records	Medium	Medium	Remove duplicate messages before training to reduce memorization
Long-message outliers	Medium	Medium	Inspect outliers before deciding whether to keep or remove them.
Noisy or unstructured text	High	High	Apply preprocessing such as lowercasing, punctuation removal, tokenization, and stopword removal.
Overlapping features	Medium	Low	Review correlation between features and avoid relying only on message length and word count.
Weak spam detection recall	Medium	High	Prioritize recall and F1-score during evaluation to reduce missed spam messages

8. Additional Requirements

8.1 Dataset or Scope Change

The project scope has remained unchanged since the initial proposal. The team continues to focus on supervised machine learning for spam and phishing detection using the selected Email Spam Detection Dataset. Although alternative datasets were briefly evaluated, the original labeled dataset was retained because it supports the project's classification objectives and machine learning requirements.

8.2 Code and Reproducibility

https://github.com/CarlosB3/Spam-Phishing-Detector-/blob/main/README.md

9. Conclusion

This progress report summarizes the work completed on the Phish Detector project through Assignments 1, 2, and 3. The team has established the project objectives, selected and analyzed an appropriate dataset, and explored the data using both statistical and visualization techniques. These analyses provided valuable insight into the dataset's characteristics, including class imbalance, duplicate records, feature relationships, and distribution patterns.

Based on these findings, the team has developed a preprocessing strategy and identified several machine learning models for future evaluation. The next phase of the project will focus on data preprocessing, feature extraction, model training, performance evaluation, and deployment through a Streamlit-based demonstration. Overall, the project remains on schedule and continues to provide a strong foundation for developing an effective spam and phishing detection system.